

Evaluating the role of pretraining dataset size and diversity on single-cell foundation model performance

Received: 13 January 2025

Accepted: 6 May 2026

Published online: 09 June 2026

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Alan DenAdel¹, Madeline Hughes², Akshaya Thoutam³, Anay Gupta⁴, Andrew W. Navia³, Nicolo Fusi², Srivatsan Raghavan^{3,5,6,7,8}, Peter S. Winter^{3,8}, Ava P. Amini^{2,8}✉ & Lorin Crawford^{2,8}✉

The success of transformer-based foundation models on natural language and images has motivated their use in single-cell biology. Single-cell foundation models have been trained on increasingly larger transcriptomic datasets, scaling from initial studies with 1 million cells to newer atlases with over 100 million cells. Here we investigate the role of pretraining dataset size and diversity on the performance of single-cell foundation models on both zero-shot and fine-tuned tasks. Using a large corpus of 22.2 million cells, we pretrain a total of 400 models, which we evaluate by conducting 6,400 experiments. Our results show that current methods tend to plateau in performance with pretraining datasets that are only a fraction of the size of current training corpora. Unlike large language models, single-cell foundation models show no clear data scaling laws, indicating that developers should focus on balancing model capacity, dataset size and computational resources rather than indiscriminately increasing all three.

The development of single-cell transcriptomic assays has empowered the dissection of biological heterogeneity at multiple scales. Machine learning models have been effectively applied to a range of tasks using single-cell data as input, including batch integration^{1,2}, imputation³, perturbation response prediction^{4,5} and reference mapping⁶. In parallel, large general-purpose neural network models, known as foundation models, have revolutionized the fields of natural language processing (NLP) and computer vision⁷. They have even led to groundbreaking progress in biological research domains such as protein modeling^{8–10}. Foundation models typically follow a transfer learning paradigm where the model is trained on a generic task, often predicting masked tokens, and then fine-tuned for specific tasks of interest⁷. Notably, recent NLP foundation models have shown strong performance in zero-shot and few-shot scenarios, where no task-specific fine-tuning is performed¹¹.

In single-cell biology, foundation models hold the potential to enable complex analyses without requiring large amounts of new data for each new scientific question. Recently, several foundation models for single-cell transcriptomics have been developed, including scBERT¹², Geneformer¹³, scGPT¹⁴ and scFoundation¹⁵. Although these models differ in their tokenization and training schemes, most rely on the transformer architecture with self-attention¹⁶. Typically trained on tens of millions of single-cell transcriptomes, these models are applied to a wide-range of tasks, including batch integration across different sequencing technologies¹⁴, cell-type annotation in previously unseen experimental datasets^{12,14,15}, inference of gene regulatory networks underlying tissue-specific mechanisms¹³ and generation of gene expression profiles to simulate transcriptomic states following exogenous perturbations^{13–15}. The ambition of these single-cell

¹Center for Computational Molecular Biology, Brown University, Providence, RI, USA. ²Microsoft Research, Cambridge, MA, USA. ³Broad Institute of MIT and Harvard, Cambridge, MA, USA. ⁴School of Computer Science, Georgia Institute of Technology, Atlanta, GA, USA. ⁵Department of Medical Oncology, Dana-Farber Cancer Institute, Boston, MA, USA. ⁶Harvard Medical School, Boston, MA, USA. ⁷Department of Medicine, Brigham and Women's Hospital, Boston, MA, USA. ⁸These authors contributed equally: Srivatsan Raghavan, Peter S. Winter, Ava P. Amini, Lorin Crawford. ✉e-mail: ava.amini@microsoft.com; lcrawford@microsoft.com

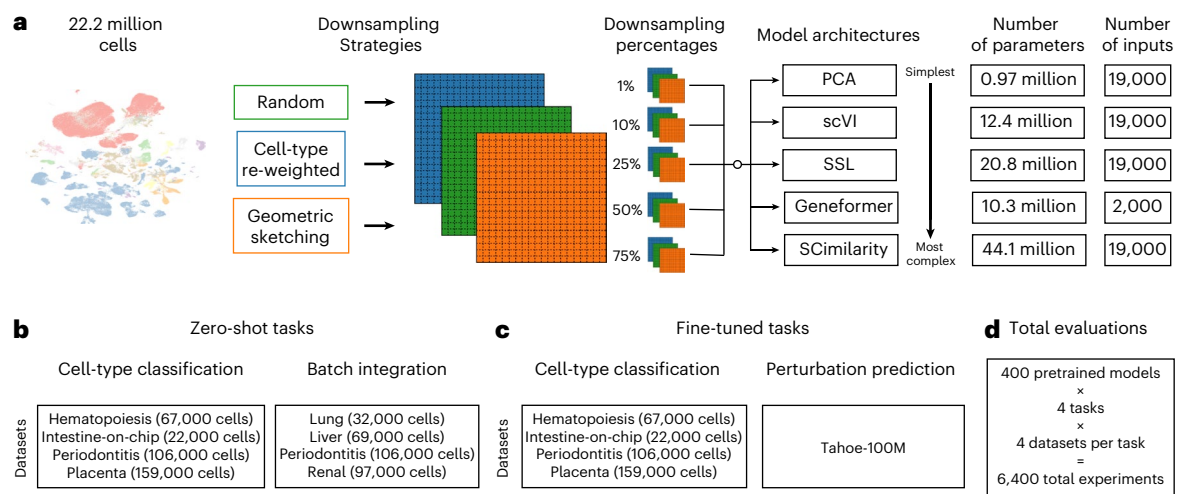


Fig. 1 | Strategy to assess the effects of pretraining dataset size and diversity on scFM performance. **a**, A schematic of the downsampling approaches, sizes of downsampled pretraining datasets and model architectures evaluated. **b**, An overview of zero-shot tasks and the datasets used for model evaluation. The zero-shot tasks include (1) cell-type classification by performing nearest-neighbor classification using model embeddings and (2) an evaluation of how well model embeddings integrate out batch effects. **c**, An overview of fine-tuned tasks and

the datasets used for model evaluation. The fine-tuned tasks include (1) cell-type classification by applying a trained classifier to the model embeddings and (2) perturbation response prediction, where a trained regression model is applied to the model embeddings. **d**, The 400 different pretrained models were evaluated on 4 different tasks, with each task consisting of 4 different datasets, yielding a total of 6,400 experiments conducted in our main analysis.

foundation models (scFMs) is to achieve state-of-the-art performance across these tasks and beyond¹⁷. While foundation models in NLP and computer vision exhibit remarkable zero- and few-shot capabilities^{11,18}, the extent to which scFMs can demonstrate analogous levels of performance remains uncertain, as they have been shown to underperform relative to simple baselines^{19–21}.

Pretraining transformer-based models typically requires expensive, specialized hardware such as graphics processing units (GPUs), tensor processing units (TPUs) or neural processing units (NPUs), as well as days to weeks of time for computation. To date, many scFMs are closed-source or only provide source code for fine-tuning and evaluation, rather than releasing the source code needed to pretrain a model from scratch. Keeping these computational and practical challenges in mind, studies in NLP have demonstrated that more data is not always better²². For a fixed training budget, curating a high-quality subset of the data and making additional training passes on that subset may yield better model performance²². Although there is some evidence suggesting that current large language models may be under-trained^{23,24}, this phenomenon has not yet been evaluated in current scFMs. Nevertheless, developers of scFMs continue to scale up both model and training dataset size. Curating datasets with increased quality and diversity, rather than focusing solely on size, could plausibly improve the performance of scFMs.

In this work, we investigate the role of pretraining dataset size and diversity on the performance of scFMs (Fig. 1). We pretrain a total of 400 models and evaluate them by conducting 6,400 experiments across a comprehensive suite of factors, including both in zero-shot scenarios and when fine-tuned for specific tasks. While many current studies on scFMs focus on assessing their abilities on downstream analyses such as gene network inference^{13,14}, model evaluations often lack ground-truth labels, making reliable comparisons difficult. Here, we focus on relatively simple, but well-defined, tasks that utilize human-curated labels for classification or aim to predict gene expression directly—namely, batch integration, cell-type annotation and perturbation response prediction. Notably, these tasks have been frequently used to evaluate scFMs because of their ease of implementation, practical utility, interpretability and presence of gold standard labels^{19–21}. Through this assessment, we interrogate the role of scaling pretraining dataset size across multiple model architectures while holding fixed both the

number of parameters within a given architecture and the training steps allocated. Our results indicate that current scFMs do not benefit from increasing pretraining dataset size or from naive approaches to increasing pretraining dataset diversity.

Results

Overview of pretraining datasets and experimental design

We evaluated the performance of five model architectures of increasing complexity: (1) projection matrices from principal component analysis (PCA) ‘pretrained’ on a given single-cell dataset, (2) a variational autoencoder (single-cell variational inference, scVI)¹, (3) a masked autoencoder (self-supervised learning, SSL)²⁵, (4) a transformer (Geneformer)¹³ and (5) an autoencoder-based metric learning approach (SCimilarity)²⁶. Detailed descriptions of the specific parameters and training procedures for each model architecture are described in Methods.

We pretrained each model architecture on datasets of varying size and diversity (Fig. 1a). These pretraining datasets were sampled from the 22.2-million-cell scTab corpus of single-cell transcriptomes²⁷. To generate these datasets, we implemented three distinct downsampling schemes: (1) random downsampling, which selects cells uniformly at random; (2) cell-type re-weighting, which attempts to include equal proportions of each cell type in the dataset; and (3) geometric sketching, which samples cells uniformly across transcriptional space without relying on metadata²⁸. Random downsampling conserves diversity (that is, all subsets have diversity that is approximately equal to the full dataset both in terms of metadata and transcriptomics), while both cell-type re-weighting and geometric sketching increase diversity relative to the full corpus. Namely, cell-type re-weighting increases diversity with respect to metadata labels, while geometric sketching increases transcriptomic-specific diversity. Pretraining datasets were generated using each of these three downsampling schemes at sizes of 1%, 10%, 25%, 50% and 75% of the total training corpus. Each pretraining corpus was split into train, validation and test sets in proportions of 80%, 10% and 10%, respectively. To ensure robust evaluation, we generated five independent splits for each combination of downsampling scheme and dataset size, resulting in 80 pretraining datasets. All five model architectures were pretrained on each of these datasets, yielding a total of $80 \times 5 = 400$ pretrained models for this initial analysis.

Downsampling schemes increase diversity in pretraining datasets

To confirm that each downsampling scheme had the intended effect on dataset diversity, we computed three diversity metrics for each pretraining dataset: (1) the Shannon Index²⁹, (2) the Gini–Simpson Index³⁰ and (3) the Vendi Score³¹. The Shannon Index and the Gini–Simpson Index evaluate diversity on the basis of metadata (in this case, the cell-type labels), while the Vendi Score directly evaluates the transcriptional diversity. Both cell-type re-weighting and geometric sketching increased diversity compared with random downsampling (Supplementary Fig. 1). Interestingly, geometric sketching resulted in higher Vendi Scores compared with cell-type re-weighting, whereas cell-type re-weighting resulted in higher Shannon and Gini–Simpson Index values. These differences reflect the underlying mechanisms of the two downsampling methods: cell-type re-weighting enhances diversity on the basis of cell-type metadata, while geometric sketching directly targets transcriptional data. The sizes for each dataset range from less than 1 gigabyte (GB) for the 1% datasets to approximately 77 GB for the 100% datasets (Supplementary Fig. 2).

To test whether increased dataset size and/or diversity improved performance (Fig. 1b), we evaluated all 400 pretrained models in the two primary regimes in which a scFM might be utilized: (1) zero-shot and (2) fine-tuned. In the zero-shot regime, the model's internal representation of data, referred to as an embedding, is used directly without additional training. By contrast, the fine-tuning regime involves re-training all or part of a model to address a specific task. Fine-tuning typically tackles a supervised learning problem where both labels and the task are well-defined, while the zero-shot approach is suited for tasks where no ground-truth labels are available.

Cell-type classification performance plateaus with limited pretraining data

A common task in many single-cell RNA-sequencing (scRNA-seq) workflows is labeling cell types on the basis of their transcriptional profiles. To evaluate initial model performance, we used all pretrained models to classify cells from a clonal hematopoiesis dataset³² in both the zero-shot scenario and after fine-tuning the models to classify cell types present in the dataset. For the zero-shot cell-type classification task, model embeddings were used directly with a nearest-neighbor classifier to classify cells. For the fine-tuned cell-type classification task, a neural network classifier was trained using the model embeddings as input.

As simple zero-shot baselines for comparison, we included embeddings derived from using highly variable genes (HVGs) directly and from the top 50 principal components. For both of these embeddings, a nearest-neighbor algorithm was used to perform classification, similar to the pretrained models. For a fine-tuned baseline, we included a regularized logistic classifier trained using HVGs as input features. In both cases, we also implemented 'non-pretrained' versions of each model using 0% of the pretraining data. To do this, we simply performed evaluations using each model's randomly initialized weights. Model performances, as assessed by the micro F1 score, are shown in Fig. 2. Interestingly, in the zero-shot case, pretrained PCA, Geneformer and SCimilarity saw no increase in performance as a function of pretraining dataset size across all downsampling schemes (Fig. 2a). scVI and SSL saw minimal improvement from increasing pretraining dataset size beyond 1–10% of the entire corpus, except in the case of geometric sketching.

When the models had been fine-tuned on the clonal hematopoiesis dataset, this pattern was amplified (Fig. 2b). Here, the previously described randomly initialized, non-pretrained models were also fine-tuned (without any pretraining). None of the pretrained PCA, SSL, Geneformer and SCimilarity showed any performance increase from the pretraining dataset size, while scVI exhibited a slight improvement in performance, going from 1% of the full corpus to 10% of the full corpus, but no improvement beyond that (except for geometric sketching, which demonstrated reduced performance overall). Notably,

we observed a trend similar to recent works^{19,20}, where the absolute classification accuracy of more complex methods such as SSL and Geneformer was exceeded by both simpler approaches (that is, pretrained PCA and scVI) and baselines (that is, PCA and logistic regression). While SCimilarity, regardless of downsampling strategy, outperformed all approaches, including the logistic regression baseline (albeit marginally in absolute terms), it still did not exhibit improvements from increasing pretraining dataset size. Model performance was comprehensively assessed using accuracy, precision, recall, micro F1 score and macro F1 score in both the zero-shot scenario (Supplementary Fig. 3) and the fine-tuning scenario (Supplementary Fig. 4). Across all metrics, the phenomenon of model performance plateauing at a small fraction of the total training data was consistently observed. Furthermore, increasing the diversity of pretraining datasets did not improve model performance in this classification task.

Saturation for the cell-type classification task is observed across datasets

To determine whether the performance plateau observed on the clonal hematopoiesis dataset generalized to other datasets, we evaluated cell-type classification in both the zero-shot and fine-tuned scenarios across several distinct datasets representing diverse biological contexts: (1) clonal hematopoiesis³² (Fig. 2), (2) intestine-on-a-chip³³, (3) placental infection (which we refer to as 'placenta')³⁴ and (4) periodontitis³⁵. Importantly, these datasets are not present in the scTab subset of CELLxGENE, and thus, represent independent test sets.

For each evaluation dataset, we sought to identify the pretraining corpus fraction where performance gains became marginal, which we refer to as the 'learning saturation point' (Fig. 3a). For each combination of model, evaluation dataset and downsampling scheme, we identified the maximum micro F1 score across all pretraining dataset sizes and defined the 'saturation threshold' as 95% of this value. We then identified the smallest percentage of the training data that produced a model surpassing this threshold and labeled this percentage the 'learning saturation point'.

Across all model architectures, downsampling strategies and evaluation datasets, zero-shot classification performance typically saturated at a small fraction of the total pretraining corpus (Fig. 3b and Supplementary Figs. 5–7), and fine-tuned performance typically saturated at 1% of the training data and never required more than 10% to reach saturation (Fig. 3c and Supplementary Figs. 8–10). For pretrained PCA, scVI and SCimilarity, zero-shot learning saturation largely occurred at 1% and never exceeded 10%; by contrast, for SSL, saturation occurred at 10% or 25% in the zero-shot case. Geneformer overwhelmingly saturated at 1% or 10%, though it reached 25% in one case. Across all four evaluation datasets, more diverse pretraining datasets did not demonstrate improved performance relative to the randomly downsampled datasets. These results suggest that, for cell-type classification, the learning saturation point for the tested scFMs can occur as early as 1% of the scTab pretraining corpus, amounting to approximately 200,000 cells. This trend holds for both transformer-based and nontransformer model architectures, as well as for both random and diversity-based downsampling schemes. Notably, 1% and 10% of the total corpus corresponds to less than 1 GB and 9 GB of raw counts, respectively (Supplementary Fig. 2).

Saturation behavior replicates for zero-shot batch integration

Next, we assessed the performance of the model embeddings on the task of batch integration (Supplementary Figs. 11 and 12). As obtaining ground-truth scRNA-seq data that has been batch-effect corrected is not possible, batch integration is inherently a zero-shot task. For this task, model embeddings were evaluated in terms of how well cells of the same type but from different batches were mixed in the embedding space. We utilized four datasets that contained cells from multiple batches, studies or assays for this evaluation including: (1) the periodontitis

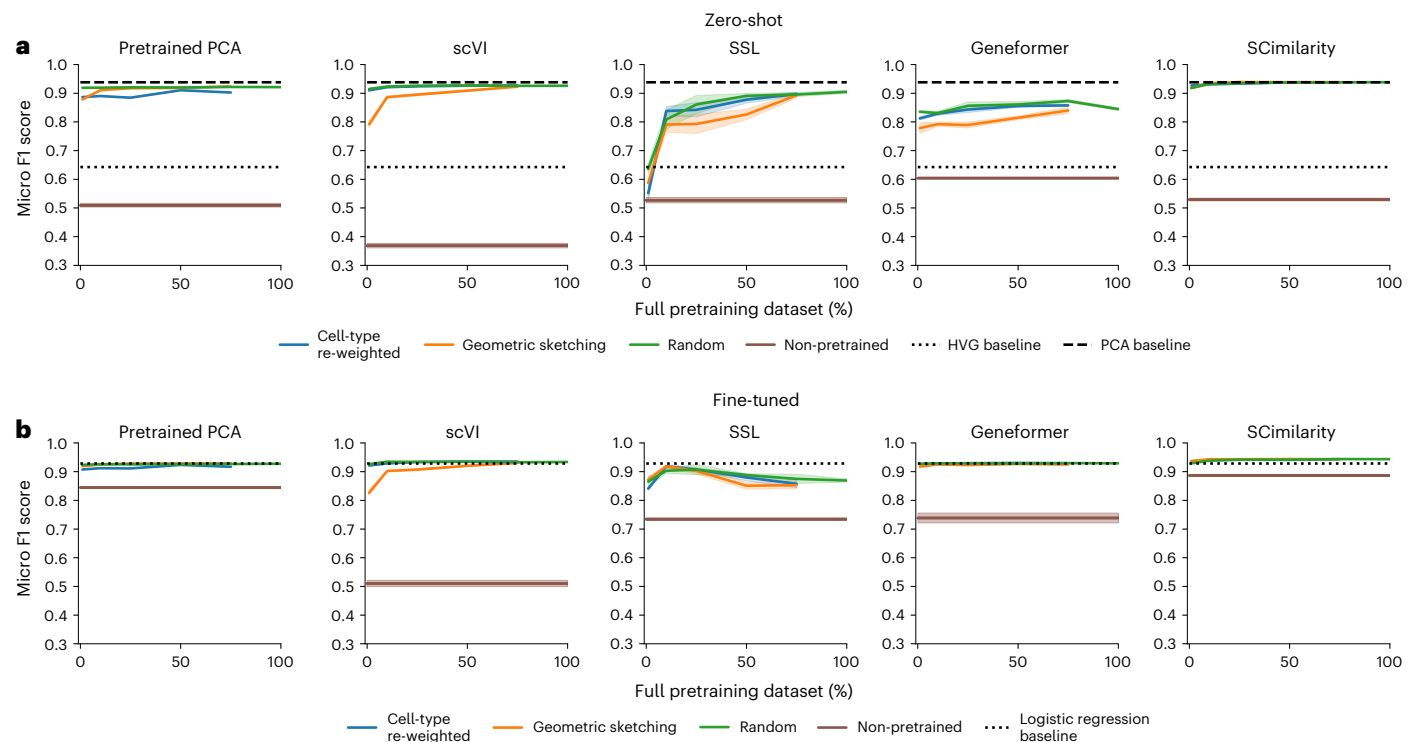


Fig. 2 | Zero-shot and fine-tuned performance on classifying cells from a clonal hematopoiesis dataset plateaus at a small fraction of the total data available for pretraining. a, Line plots showing zero-shot classification performance for each model's embeddings, as evaluated by the micro F1 score. For each model, the different colors correspond to the downsampling strategy used to generate the data used for pretraining. The dotted line shows the performance of using the HVGs as an embedding, the dashed line shows the performance of using principal component projections as an embedding and the solid brown line corresponds to the 'non-pretrained' version of each model, where evaluations were done using the randomly initialized weights. The shaded area represents the standard error across five independent splits. **b,** Line plots showing the classification

performance for each model after fine-tuning, as evaluated by the micro F1 score. For each model, the different colors correspond to the downsampling strategy used to generate the data used for pretraining. The dotted line shows the performance of training a regularized logistic regression classifier using the HVGs as input features, and the brown solid line corresponds to the 'non-pretrained' version of each model, where evaluations were done using the randomly initialized weights. In the fine-tuned tasks, this means that a randomly initialized model is frozen and fine-tuned on the evaluation dataset but with no pretraining (as opposed to having no training at all). The shaded area represents the standard error across five independent splits.

dataset³⁵, previously used in the cell-type classification evaluation; (2) a lung dataset³⁶; (3) a renal dataset³⁷; and (4) a liver dataset³⁸.

We adopted the batch integration evaluation methodology from the studies by Kedzierska et al.¹⁹ and Luecken et al.³⁹. This involved generating cluster assignments on the basis of Louvain clusters obtained from the embedding space and comparing them with the ground-truth cell-type labels. The quality of these cluster assignments was assessed using normalized mutual information (NMI) and Adjusted Rand Index (ARI). To evaluate the degree of batch mixing in the embedding space, we calculated the average silhouette width (ASW) with respect to both cell-type labels and batch labels. The average BIO (AvgBIO) score was then used to summarize overall performance across these metrics, combining ASW, NMI and ARI. The AvgBIO results for the lung dataset are presented in Fig. 4a, and all metrics for the periodontitis, lung, liver and renal datasets are shown in Supplementary Figs. 11–14. As observed with cell-type classification, the absolute performances of SSL and Geneformer were often exceeded by both simpler approaches (that is, pretrained PCA and scVI) and baselines (that is, using HVGs as embeddings and PCA). The one exception here was, again, SCimilarity, which showed a similar performance to scVI.

We repeated the learning saturation point analysis for the AvgBIO batch integration performance curves across all models, evaluation datasets and downsampling schemes, and observed that, consistent with the classification task analysis, model performance generally saturated at a small fraction of the total pretraining corpus (Fig. 4b). Specifically, the saturation of batch integration performance occurred

at 1% for pretrained PCA, between 1% and 10% for scVI and SCimilarity, 0% and 50% for SSL, and between 1% and 50% for Geneformer. Notably, in the renal dataset, the 'non-pretrained' SSL models performed on par with all of the trained models (Supplementary Fig. 14).

Saturation behavior replicates for fine-tuned perturbation response prediction

Given the saturation behavior observed on both cell-type classification and batch integration, we next assessed the performance of the model embeddings on perturbation response prediction as a final downstream task (Fig. 5). Although obtaining ground-truth scRNA-seq perturbation data for individual cells is not possible, we fine-tuned models on pairs of perturbed and unperturbed cells from the same cell lines (Methods). For this task, we assembled a subset of the large Tahoe-100M experimental atlas⁴⁰. Specifically, we used all cells that were from the (1) LoVo, (2) PANC-1, (3) NCI-H2030 and (4) BT-474 cell lines that were each treated with one of four different drugs: (1) homoharringtonine, (2) dinaciclib, (3) PH-797804 and (4) TAK-901, or a control treatment of dimethyl sulfoxide (DMSO). To conduct this analysis, we first pretrained each of PCA, scVI, SSL, Geneformer and SCimilarity on a downsampled proportion of cells from scTab. During fine-tuning, we then took the model embedding corresponding to an unperturbed cell and trained a simple multilayer perceptron (MLP) to predict the gene expression of a real perturbed cell of the same type, randomly selected from Tahoe-100M. For each drug, we fine-tuned on three of the four cell lines and left one out for the evaluation. Specifically, we

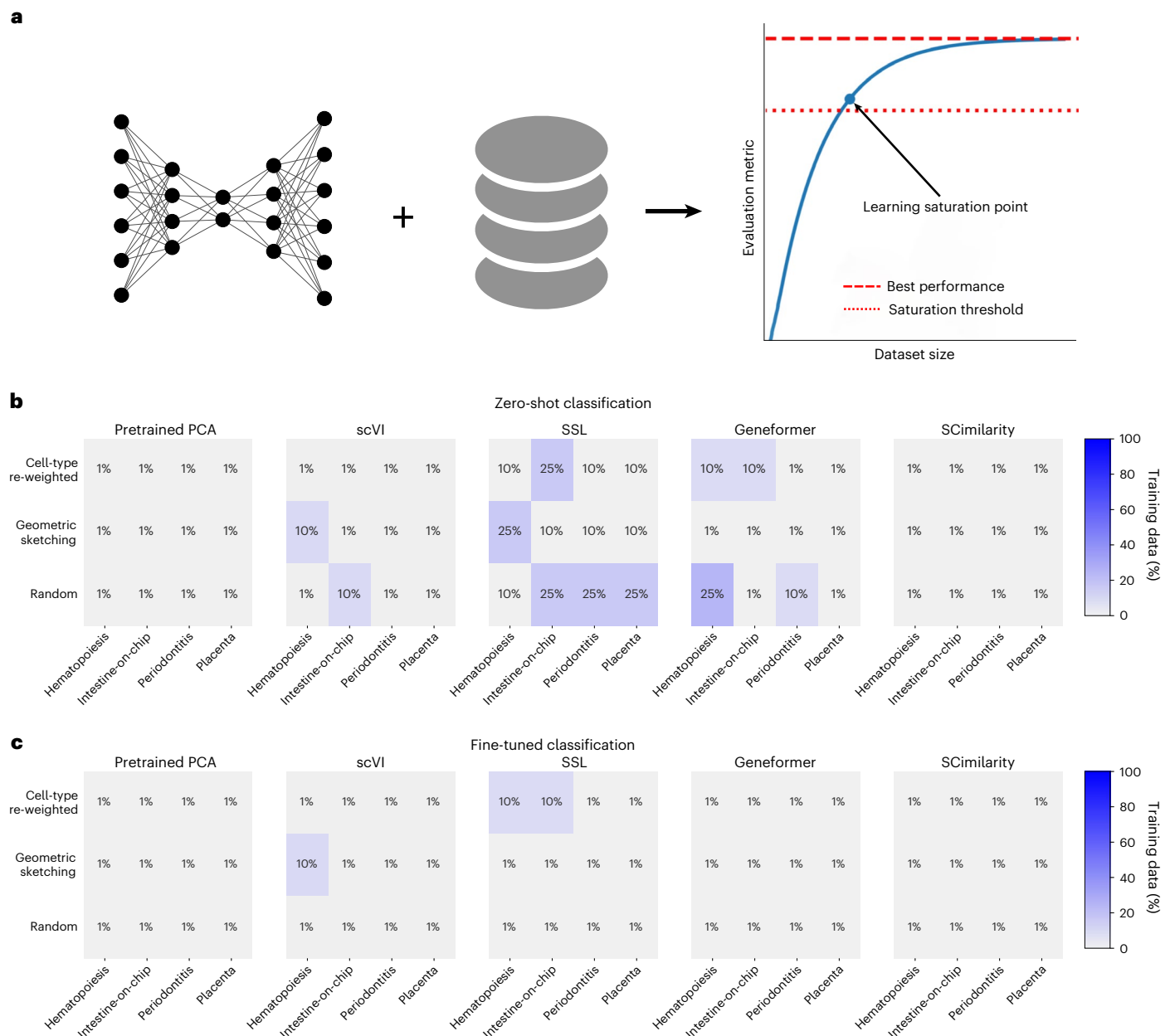


Fig. 3 | Performance saturation as a function of pretraining dataset size across multiple evaluation datasets on zero-shot and fine-tuned cell-type classification. **a**, A schematic of the analysis to find the learning saturation point. For each family of models (that is, a downsampling strategy paired with a model), a 'saturation threshold' of 95% of the maximum performance was computed, and the minimum pretraining dataset size that produced a model surpassing that threshold was identified. This dataset size was denoted the 'learning saturation point' and is considered the point at which model performance saturated as a function of pretraining dataset size. **b**, A heat map visualizing the learning saturation point for the clonal hematopoiesis, intestine-on-chip, periodontitis and placental infection datasets for each of pretrained PCA,

scVI, SSL, Geneformer and SCimilarity across each downsampling strategy and when evaluated in the zero-shot regime. Each sub-panel corresponds to the model architecture, the x-axis corresponds to the dataset evaluated and the y-axis corresponds to the downsampling strategy used to pretrain each model. **c**, A heat map visualizing the learning saturation point for the clonal hematopoiesis, intestine-on-chip, periodontitis and placental infection datasets for each of pretrained PCA, scVI, SSL, Geneformer and SCimilarity across each downsampling strategy and when evaluated in the fine-tuning regime. Each sub-panel corresponds to the model architecture, the x-axis corresponds to the dataset evaluated and the y-axis corresponds to the downsampling strategy used to pretrain each model.

held out LoVo for homoharringtonine, NCI-H2030 for dinaciclib, BT-474 for PH-797804 and PANC-1 for TAK-901. Each model was assessed by computing the R^2 between the average expression of each individual gene in the actual perturbed cells and the corresponding predicted expression vectors for the same cell type. We also computed the mean squared error (MSE) between the gene expression values in the actual perturbed cells and the predicted expression. The R^2 performance curves for predicting perturbation effects on the LoVo cell line treated

with homoharringtonine are shown in Fig. 5a. The results for all other drugs across the other held out cell lines are similar whether evaluated by either R^2 or MSE (Supplementary Figs. 15–18).

We repeated the learning saturation point analysis for the R^2 performance curves (chosen because it ranges between 0 and 1, in line with the metrics for other tasks) across all models, evaluation datasets and downsampling schemes. Consistent with the classification task analyses, we observed that model performance generally saturated at a

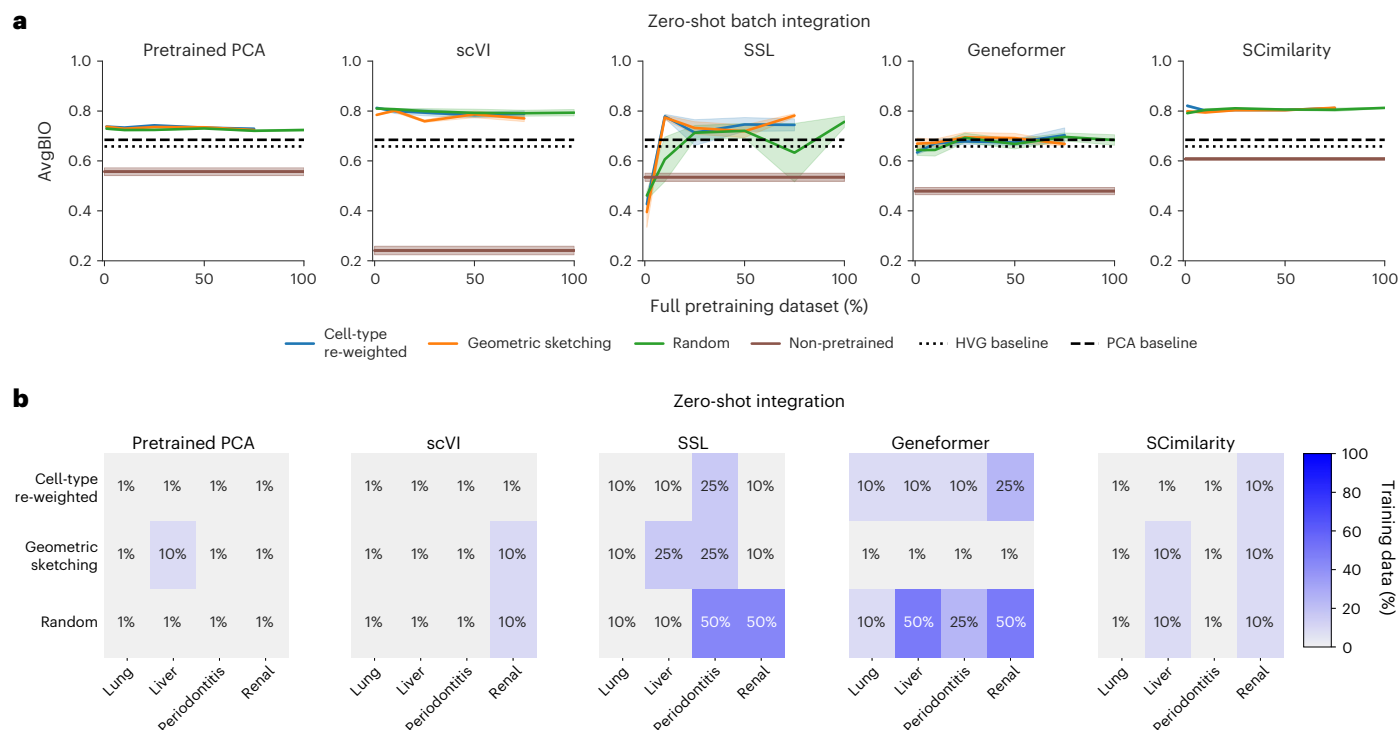


Fig. 4 | Performance saturation as a function of pretraining dataset size across multiple evaluation datasets on zero-shot batch integration. a, Line plots showing zero-shot batch integration performance for each model's embeddings, as evaluated by the AvgBio metric on the lung dataset from Kim et al.³⁶. For each model, the different colors correspond to the downsampling strategy used to generate the data used for pretraining. The dotted line shows the performance of using the HVGs as an embedding, the dashed line shows the performance of using principal component projections as an embedding and the brown solid line

corresponds to the 'non-pretrained' version of each model, where evaluations were done using the randomly initialized weights. The shaded area represents the standard error across five independent splits. **b,** A heat map visualizing the learning saturation point for the lung, liver, periodontitis and renal datasets for each of pretrained PCA, scVI, SSL, Geneformer and SCimilarity across each downsampling strategy. Each sub-panel corresponds to the model architecture, the x axis corresponds to the dataset evaluated and the y axis corresponds to the downsampling strategy used to pretrain each model.

small fraction of the total pretraining corpus (Fig. 5b). Specifically, the saturation of perturbation response prediction performance occurred at 1% of the pretraining dataset size for all model architectures and across all datasets. In line with recent results evaluating deep neural network perturbation response prediction performance²¹, for all drugs except dinaciclib (Supplementary Fig. 16), a simple baseline of predicting 'no change' in gene expression outperformed each of the fine-tuned models (Supplementary Figs. 15, 17 and 18).

Assessing the impact of including perturbation data in the pretraining corpus

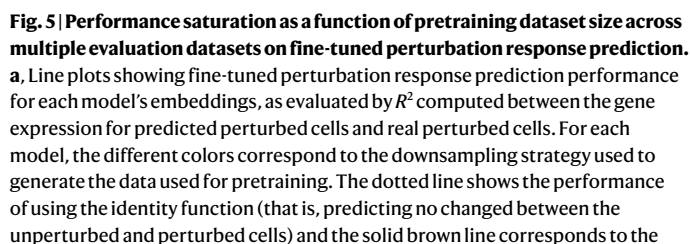
As increasing the diversity of the pretraining datasets did not improve model performance on downstream tasks, we reasoned that a small fraction (at maximum, 25%) of the scTab corpus was sufficient to represent cells with a normal transcriptional phenotype. We thus hypothesized that supplementing the pretraining corpus with cells that had been systematically perturbed could enhance the contribution of pretraining on downstream performance.

Here, we performed two sets of spike-in experiments using genome-wide Perturb-seq data from Replogle et al.⁴¹, which contains over 2.5 million systematically perturbed cells. For both sets of experiments, we started with the uniformly downsampled scTab pretraining datasets. For each of these datasets, we generated two new datasets by 'spiking-in' cells from the Replogle Perturb-seq experiments. This involved replacing a portion of the scTab cells with an equal number of cells from the Replogle Perturb-seq data. The spike-in experiments were conducted at two proportions of spike-in perturbation data: 10% and 50%. The Replogle dataset had enough cells to replace 10% of the scTab cells in all of the uniformly downsampled datasets. However,

there were only enough cells to replace 50% of the scTab cells in the 1%, 10% and 25% downsampled datasets. These experiments allowed us to assess whether incorporating perturbed cells could increase the value of pretraining for downstream tasks. We used these 'spike-in' datasets to pretrain scVI, SSL and Geneformer as representative approaches of a variational autoencoder, a masked autoencoder and a transformer.

To evaluate the effect of adding in perturbation data, we analyzed the learning saturation points of the micro F1 score performance curves for the cell-type classification task, evaluating all models in both the zero-shot (Supplementary Fig. 19) and fine-tuning (Supplementary Fig. 20) cases. All metrics are shown for the zero-shot evaluations in Supplementary Figs. 21–24 and for the fine-tuned evaluations in Supplementary Figs. 25–28. Consistent with the initial analysis of the classification task, model performance generally saturated at a small fraction of the total pretraining corpus. In the fine-tuned case, all model architectures typically saturated at 1%, although Geneformer saturated at 25% for the 10% spike-in in one instance. For scVI, saturation in the zero-shot case typically occurred at 1% and never exceeded 10%. By contrast, SSL saturation occurred between 10% and 25% in the zero-shot case. Finally, Geneformer typically saturated at 1% or 10% in the zero-shot case.

We again repeated the learning saturation point analysis on the AvgBio performance curves for the batch integration task, evaluating all models pretrained on the spike-in datasets (Supplementary Fig. 29). All metrics are shown for the zero-shot batch integration evaluations in Supplementary Figs. 30 and 31. Similar to the classification tasks, across all combinations of model, evaluation dataset and downsampling schemes, model performance generally saturated at a small fraction of the total pretraining corpus, even with the inclusion of spike-in data.



‘non-pretrained’ version of each model, where evaluations were done using the randomly initialized weights. The shaded area represents the standard error across five independent splits. **b**, A heat map visualizing the learning saturation point for each perturbation dataset across each downsampling strategy when evaluated on fine-tuned perturbation response prediction. Each sub-panel corresponds to the model architecture, the x axis corresponds to the dataset evaluated and the y axis corresponds to the downsampling strategy used to pretrain each model.

First, to evaluate the role of model size (that is, the number of parameters in a model), we chose two representative architectures: scVI (representing autoencoders) and Geneformer (representing transformer-based methods). For scVI, we trained models with 0.5 $\times$, 2 $\times$ and 4 $\times$ the number of parameters relative to the default implementation by increasing both the size of the two hidden layers and the dimensionality of the latent embedding space. Counting the total number of parameters in Geneformer is more complicated; so, instead, we simply downsized the latent embedding space from the default 256 dimensions to 32, 64 and 128 dimensions. When evaluating scVI, we

We also evaluated these models of various sizes on the task of zero-shot batch integration. By contrast to the general result that increasing model scale generally improved zero-shot cell-type classification performance, we observed that for zero-shot batch integration, scVI (Fig. 6c) sometimes showed a better performance with a larger model (for example, the renal dataset), but typically larger models performed worse (for example, on the lung, periodontitis and liver datasets). Notably, they were still outperformed by the PCA baseline in the periodontitis, renal and liver datasets. For Geneformer, the performance on batch integration was quite similar to cell-type classification, where the two larger model sizes showed improved performance; still, however, a latent dimension of 256 did not show benefits over a latent

dimension of 128 (Fig. 6d). Further, pretraining dataset size had no impact on model performance across all model scales for both scVI and Geneformer.

For Geneformer, the performance on batch integration was quite similar to cell-type classification and was worse in absolute terms than scVI, reinforcing the observations that Geneformer has poor scaling properties as a function of dataset size and model size (Figs. 2–6). Experiments varying the weight decay hyperparameter had little to no impact on the zero-shot classification (Supplementary Fig. 32a) or zero-shot batch integration (Supplementary Fig. 32b) tasks, underscoring the robustness of these results.

Last, as scVI and SSL are both generic autoencoders that use different loss functions, we additionally trained a model we refer to as an ‘SSL VAE’ that integrates the variational loss term of scVI with the masked training objective of SSL. This combined loss function proved more difficult to train, with many models not fully converging in the same number of steps as the original SSL implementation. This resulted in poor performance for zero-shot cell-type classification tasks (Supplementary Fig. 33).

Together, these results demonstrate that, across the autoencoder and transformer architectures, larger models still saturate with small pretraining datasets, although they exhibit better overall performance relative to small models.

Discussion

In this study we investigated the role of pretraining dataset size and diversity on the performance of general-purpose scFMs. We evaluated five representative general-use models for scRNA-seq data on downstream tasks in both the zero-shot case and after fine-tuning the models for specific tasks. Our work addresses a critical consideration in training large-scale models: the size and diversity of the pretraining corpus. While neural scaling laws observed in other domains suggest that increasing dataset size leads to better model performance, our findings demonstrate that, past a learning saturation point, simply increasing pretraining datasets, even with greater diversity, does not necessarily improve the performance of scFMs on downstream tasks. These results suggest that further scaling-up of pretraining datasets from tens of millions of cells to hundreds of millions or even billions of cells may not yield tangible returns.

Importantly, our analysis used the scTab dataset as-is, without additional quality control or data curation beyond downsampling. Recent work has highlighted that some reference atlases contain substantial numbers of low-quality cells and contaminants⁴². Our findings underscore the importance of prioritizing data quality and content over sheer dataset size. This observation is similar to what has been shown in recent work, which suggests that performance gains in scFMs can be achieved through principled optimization of smaller training corpora, for example, by combining single-cell datasets that capture highly variable, yet unique, transcriptomic states⁴³. Our experiments showing that spiking-in perturbation data does not necessarily improve performance on cell-type classification or batch integration additionally emphasize the importance of using pretraining data and a pretraining task that are aligned with the downstream tasks for which a

model will be applied. As such, developers of scFMs and large databases should consider data quality and content rather than simply attempting to gather the largest datasets possible, which we have shown is unlikely to improve performance meaningfully. This is especially important as current foundation models are also quite computationally expensive for both pretraining and inference (Supplementary Fig. 34). Still, a key limitation of our study is that it only investigates a small number of scFMs, while there continues to be dozens of newly proposed architectures and modeling strategies^{12,14,15,44–49}.

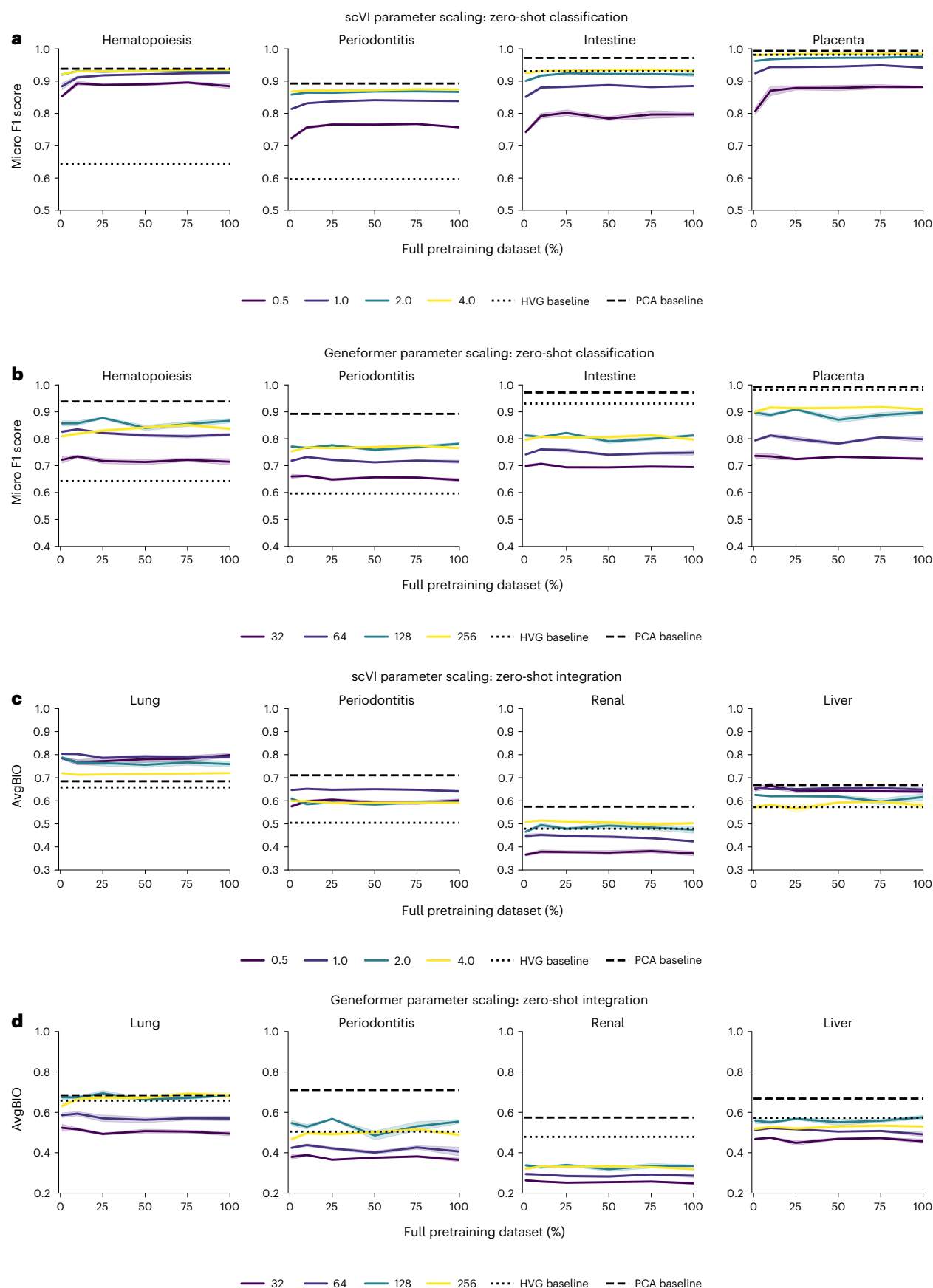
Taken as a whole, our results highlight the need for a more nuanced approach: one that balances dataset size and diversity with careful attention to model architectures, training strategies and benchmarking. While conceptualizing a cell as ‘a bag of RNA’⁵⁰ is a useful starting point for many models, the concept of a cell as ‘a sentence of genes’^{12–15} may be less effective. Well-designed embeddings have long been used to great success in transcriptomics^{1,51}, and simple embeddings can capture a large amount of the information present in single-cell data^{19,52}. Recent work has shown that aligning a model’s training objective with its intended downstream task (for example, through biologically informed loss functions) can also be an efficient way to enhance its capabilities³³. In our study, SCimilarity was the best performing model (in absolute terms) at cell-type classification, and its pretraining task is heavily aligned with representation learning of cell types²⁶. The same holds true for the batch integration task, where SCimilarity’s contrastive loss function implicitly teaches the model to embed cells of the same type together regardless of batch, reinforcing the idea that aligning the training objective with the downstream task of interest is important. Although transformer-based models have shown great promise, there is a growing body of work^{19–21} showing that simple methods often outperform transformer-based models applied to scRNA-seq data. Our results emphasize this fact on multiple tasks and datasets. This suggests that current scFM tokenization schemes that produce ‘cellular sentences’ may be limited in their ability to represent scRNA-seq data faithfully.

It is worth pointing out that an interesting set of alternative single-cell modeling approaches leverage large language models directly, rather than adapting different architectures to transcriptomic data. For example, GenePT uses GPT-3.5 to generate gene-level embeddings from the National Center for Biotechnology Information’s text descriptions and then generates single-cell embeddings by averaging these gene embeddings weighted by each gene’s expression level⁴⁵. As a different strategy, Cell2Sentence adapts existing large language models for use in transcriptomics by fine-tuning them with ‘cell sentences’ consisting of cellular metadata and ranked gene names⁴⁷. Multiple architectural paths should be pursued to find an effective path forward for this class of scientific foundation models.

Though our results show that performance for many methods does not scale with additional data, a complete study over parameter counts and training budget would be required to establish whether scaling laws exist for scFMs as they do for large language models. Future work should focus on understanding the interplay between feature representation, model architecture, inference and pretraining dataset size to optimize performance on downstream tasks. Our

Fig. 6 | Assessing the impact of model size on zero-shot cell-type classification performance. Evaluations were done using the clonal hematopoiesis, intestine-on-chip, periodontitis and placental infection datasets. **a**, Line plots showing zero-shot classification performance for scVI models with varying numbers of parameters, as evaluated by the micro F1 score. For each model, the different colors correspond to the scaling factor for the total number of parameters relative to the default implementation by scaling both the size of the hidden layers and the dimensionality of the latent embedding space. By default, scVI has a latent embedding space size of 10 and a hidden layer size of 128. In all cases, scVI has one hidden layer in the encoder and one hidden layer in the decoder. The shaded area represents the standard error across five independent splits.

b, Line plots showing zero-shot classification performance for Geneformer models with varying numbers of parameters, as evaluated by the micro F1 score. For each model, the different colors correspond to the dimensionality of the latent embedding space; by default, Geneformer has an embedding size of 256. The shaded area represents the standard error across five independent splits. **c**, Similar to **a** but showing the performance of scVI models with varying numbers of parameters on the zero-shot batch integration task as evaluated by the AvgBIO metric. **d**, Similar to **b** but showing the performance of Geneformer models with varying numbers of parameters on the zero-shot batch integration task as evaluated by the AvgBIO metric.



initial results, which examine the impact of increasing the number of parameters in a model, suggest an interesting (but potentially limiting) path forward. Indeed, large language models have scaling laws that have been established with respect to data, training compute and model size⁵⁴. Even more, there have been studies focused on developing compute-optimal models that have optimal performance for a given computational budget for training. These frameworks provide guidelines such as ‘for every doubling of model size the number of training tokens should also be doubled’⁵⁵. Another potential avenue for improved data efficiency use, particularly in the regime of scRNA-seq studies where certain cell types or states are naturally more prevalent than others, is active learning via adaptive sampling^{56,57}. In this setting, algorithms improve performance by refining which samples are represented across successive training iterations, guided either by biologically informed rules or by patterns uncovered within their learned latent embeddings. Still, the architectures of current scFMs continue to grow in complexity without a complete understanding of where along the compute–data–parameter frontier a given model lies. Although there has been research showing that for scFMs, data quality has a scaling law⁵⁸, there has been no comprehensive investigation into computational, data and parameter size scaling laws for these models. Bridging this gap could unlock principled design strategies for both model architecture development and experimental data collection, as well as accelerate progress toward more generalizable and efficient models for future single-cell applications.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41592-026-03120-y>.

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Methods

Pretraining data and downsampling schemes

The main pretraining dataset used in this study is scTab²⁷, which is a subset of the CELLxGENE⁵⁹ census (version 2023-05-15). In this particular dataset, there are a total of 22.2 million cells from 164 unique cell types, 5,052 unique human donors and 56 human tissues. For each of the downsampling strategies described below (uniform random downsampling, cell-type re-weighted downsampling and geometric sketching downsampling), cells in scTab were downsampled in percentages of 1%, 10%, 25%, 50% and 75%, with five replicates for each percentage. Each of these downsampled datasets was then divided into training, validation and test sets in proportions of 80%, 10% and 10% splits, respectively. We also downsampled the entire scTab dataset (denoted as the 100% dataset) into 80% training, 10% validation and 10% test sets with five replicates of this split.

Uniform downsampling. Uniform downsampling is the simplest scheme used. The uniformly downsampled datasets sampled each cell uniformly at random without consideration for any additional metadata or the expression counts matrix.

Cell-type re-weighted downsampling. The goal of the cell-type re-weighted scheme is to downsample such that each cell type is present in equal proportions in the resulting dataset. Let $p = (p_1, \dots, p_K)$ be the proportion of each cell type in the scTab dataset where K is the total number of cell types. For many cell types present in the scTab dataset, there are not enough cells to satisfy this goal even when downsampling to 1%. Therefore, to generate a cell-type re-weighted dataset, we first choose all cells that belong to a cell type that is not common enough to satisfy the equal proportion condition. Then, we randomly down-sample among the remaining cell types with equal proportions. Let N be the total number of cells and D be the downsampling proportion. The algorithm is as follows:

1. For any k th cell type satisfying $p_k ND < ND/K$ (that is, it is impossible to satisfy the re-weighting condition, even by selecting all of the cells of this cell type), include all of the cells of this cell type.
2. Let N' be the number of unselected cells after the previous step and K' be the number of remaining cell types. Some of these cell types that previously satisfied the re-weighting condition may no longer do so with N' and K' . Repeat step 1 with N' in place of N and K' in place of K .
3. Let M be the number of cells already selected. When all remaining cell types satisfy the re-weighting condition, sample $(DN - M)/K'$ cells from each cell type (that is, re-weight these cell types to have equal proportions).

The resulting downsampled dataset should have proportions $p^* = (p_1^*, \dots, p_K^*)$ such that $p_1^* \approx p_2^* \approx \dots \approx p_K^* \approx 1/K$.

Geometric sketching. Geometric sketching is a method that down-samples on the basis of the intrinsic diversity of a dataset without reference to any metadata²⁸. Geometric sketching samples uniformly over transcriptional space without regard to density (that is, a cell type that has a large number of cells with very similar gene expression will be downweighted in the resulting 'sketch'). Let N be the total number of cells, D be the downsampled proportion and C be the number of covering hypercubes. The geometric sketching algorithm is as follows:

1. Generate a 'plaid covering' of the dataset in transcriptomic space. A plaid covering is a covering of the dataset consisting of equally sized hypercubes. Each hypercube in the covering will contain a different number of cells.
2. For each covering hypercube, sample DN/C cells.

Owing to the fact that each hypercube is of equal size, higher-density regions of transcriptomic space (that is, regions with many

similar cells) will be downweighted in the resulting 'sketch' of the dataset.

Spike-in experiments with perturbation data

To assess the extent to which including only normal transcriptomic phenotypes impacts downstream model performance, we performed two sets of 'spike-in' experiments using the Replogle et al.⁴¹ Perturb-seq study (CRISPR-based screens with scRNA-seq readouts) archived in the scPerturb database⁶⁰. The Replogle dataset contains 2,547,877 cells and 10,117 genes. For both sets of experiments, we began with the downsampled datasets generated using the uniform downsampling scheme. Then, for each of these datasets, we generated two new datasets that had cells from the Replogle Perturb-seq experiments 'spiked-in'. This was done by removing some of the scTab cells and replacing them with an equal number of cells from the Replogle study. We spiked-in these cells at rates of 10% and 50%. The Replogle dataset had enough cells to replace 10% of the scTab cells in all of the uniformly downsampled datasets; however, there were only enough cells to replace 50% of the scTab cells in the 1%, 10% and 25% downsampled datasets.

Evaluation datasets

Below is a description of each dataset we used for evaluation in this study. Recall that the clonal hematopoiesis, placenta, intestine-on-chip and periodontitis datasets were all used for zero-shot and fine-tuned cell-type classification tasks; the lung, renal and liver datasets were used to evaluate zero-shot batch integration capabilities; and the Tahoe-100M atlas was used to assess perturbation response prediction accuracy. All these data, except the lung dataset and Tahoe-100M, were downloaded from the CELLxGENE census (where each of these studies were selected because they published after the 15 May 2023 release and ensured no 'leakage' with the training corpora from scTab). The lung dataset was taken from the Curated Cancer Cell Atlas. Information on where to download these resources can be found in the Code availability and Data availability sections.

Clonal hematopoiesis. The clonal hematopoiesis data results from profiling peripheral blood gene expression from 17 patients with clonal hematopoiesis and seven controls³². For the zero-shot cell-type classification evaluations, the data were divided into 50/50 train/test splits. For the fine-tuning evaluations, the data were divided into 80/10/10 train/test/validation splits. There are 66,985 total cells and eight labeled cell types in this dataset.

Placenta. The placenta data results from profiling placental responses to three pathogens³⁴. For the zero-shot cell-type classification evaluations, the data were divided into 50/50 train/test splits. For the fine-tuning evaluations, the data were divided into 80/10/10 train/test/validation splits. In this dataset, there are 20 donors (counting both mother and fetus), 158,978 total cells and seven labeled cell types.

Intestine-on-chip. The intestine-on-chip data results from profiling induced pluripotent stem cell-derived small intestine-on-chips that were exposed to either: (1) different media conditions or (2) interferon-beta or interferon-gamma³³. For the zero-shot cell-type classification evaluations, the data were divided into 50/50 train/test splits. For the fine-tuning evaluations, the data were divided into 80/10/10 train/test/validation splits. There are 22,280 total cells and eight labeled cell types.

Periodontitis. The periodontitis data results from profiling the periodontium of 34 individuals³⁵. For the zero-shot cell-type classification evaluations, the data were divided into 50/50 train/test splits. For the fine-tuning evaluations, the data were divided into 80/10/10 train/test/validation splits. There are 105,918 total cells and 27 labeled cell types.

Lung. The lung data results from profiling metastatic lung adenocarcinoma samples from 14 different donors³⁶. Note that we do not use training and test splits for the zero-shot batch integration task. Instead, we carry out evaluations using the embeddings from each model directly. There are 32,493 total cells and ten labeled cell types.

Renal. The renal data results from profiling renal cortex samples from 19 different donors³⁷. Note that we do not use training and test splits for the zero-shot batch integration task. Instead, we carry out evaluations using the embeddings from each model directly. There are 97,125 total cells and 35 labeled cell types.

Liver. The liver data results from profiling the pediatric and adult human liver samples from 16 different donors (7 pediatric and 9 adult)³⁸. Note that we do not use training and test splits for the zero-shot batch integration task. Instead, we carry out evaluations using the embeddings from each model directly. There are 69,032 total cells and 19 labeled cell types.

Tahoe-100M. The Tahoe-100M is a single-cell atlas of approximately 100 million transcriptomic profiles resulting from 1,100 small molecule perturbations across 50 cancer cell lines⁴⁰. For our analysis, we subset the 1,100 small molecules to consider the four with the greatest median E-distance between perturbed and control cells of the same cellular identity at 5 μ M, the maximum dosage used in the study (see fig. 4 in Zhang et al.⁴⁰ for reference). We then selected four cell lines where at least one of these drugs produced a noticeable non-zero effect (again based on E-distance): PANC-1 for dinaciclib, BT-474 for homoharringtonine, NCI-H2030 for PH-797804 and LoVo for TAK-901. For each drug, we fine-tuned using the cell line with the strongest perturbation response and two of the remaining three cell lines, while holding out the fourth for evaluation. Specifically, for dinaciclib, the held out cell line was NCI-H2030, for homoharringtonin, the held out cell line was LoVo, for PH-797804, the held out cell line was BT-474, and for TAK-901, the held out cell line was PANC-1. For perturbation response evaluations, the three cell lines used for fine-tuning were divided into 90/10 train/validation splits. In our analyses, control (that is, unperturbed) cells were cells treated with DMSO. There were far more control cells than treated cells in each dataset, so we additionally subsetting down to 20,000 control cells in the training data and 2,000 cells in the validation data.

Pretraining of machine learning models

In this study, we implemented five machine learning methods for single-cell transcriptomics: PCA, scVI¹, SSL²⁵, Geneformer¹³ and SCimilarity²⁶. For each model class, we standardized the amount of compute (in terms of training steps) that was allocated to pretraining so that simply increasing the pretraining dataset size did not increase the amount of training performed. To determine the number of training epochs allocated to each model, we split the entire scTab corpus into an 80/10/10 split of train, test and validation datasets, respectively. We then trained each model class on the training split until the validation loss had converged. Next, we plotted the validation loss as a function of the number of epochs and selected the number of epochs that resulted in convergence of the loss function. Finally, this value was normalized on the basis of the fraction of training data used to determine the number of epochs needed for each training split. For example, if the number of epochs taken to converge on the full corpus was 40 and we want to find the number of epochs for 25% of the corpus, then we would use $40/0.25 = 160$ epochs for this split of the data. In addition to pretraining on corpora generated by each of the downsampling strategies described above (uniform random downsampling, cell-type re-weighted downsampling and geometric sketching downsampling) at percentages of 1%, 10%, 25%, 50% and 75%, we also implemented ‘non-pretrained’ baseline versions of each model (that is, using 0%

of the pretraining data). To do this, we simply performed evaluations using each model’s randomly initialized weights.

Pretrained PCA. To create a ‘pretrained’ PCA approach, we first subset each pretraining dataset to just include HVGs found by using the scanpy package, and we then computed the top 50 principal components of the resulting matrix. For all tasks, each evaluation dataset was projected onto these 50 principal components, and the corresponding vectors were used as embeddings.

scVI. The default parameters were used to designate the architecture of the pretrained scVI models. In particular, the number of nodes per hidden layer was 128, the latent embedding space had ten dimensions, the encoder and decoder both have a single hidden layer, the dropout rate was 10%, the gene likelihood was zero-inflated negative binomial, the dispersion parameter of the negative binomial distribution was constant per gene and the latent distribution was normal. For training, the batch size was 128, early stopping with respect to the validation loss was turned on and the number of epochs was determined as $40 \times (100/P)$, where P is the training proportion, as the number of epochs to converge on the full corpus was 40. For example, for a 25% split, the number of epochs is $40 \times (100/25) = 160$ epochs.

SSL. The parameters specified in the SSL repository on GitHub were used to specify the model architecture and pretraining parameters (except for the number of epochs, which was determined as described above). In particular, the hidden layer sizes were (512, 512, 256 and 256) for the encoder (reversed for the decoder), and the latent embedding space dimension was 64. For training, the batch size was 16,384, the masking rate was 0.5, the dropout rate was 0.1, the weight decay was set to 0.01, the max steps was 117,000 and the learning rate was 0.001.

Geneformer. The parameters specified in the Geneformer repository on Hugging Face were used to specify the Geneformer architecture and pretraining parameters (except for the number of epochs, which was determined as described above with three epochs used for the 100% datasets). In particular, the model type was bidirectional encoder representations from transformers, the maximum input size was 2,048, the number of layers was six, the number of attention heads was four, the number of embedding dimensions was 256, the intermediate size was 512 (that is, twice the number of embedding dimensions) and the activation function was ReLU. For training, the batch size was 18 per GPU (across 8 GPUs in total), the maximum learning rate was 0.001, the learning schedule was linear, the number of warm-up steps was 10,000, the attention dropout was 0.02, the hidden layer dropout was 0.02, the optimizer was AdamW, the layer norm epsilon was 1×10^{-12} , the initializer range was 0.02 and the weight decay parameter was 0.001.

SCimilarity. The parameters specified in the SCimilarity repository on GitHub were used to specify the model architecture and pretraining parameters (except for the number of epochs, which was determined as described above). In particular the hidden layer sizes were (1,024, 1,024 and 1,024) for the encoder (reversed for the decoder), the size of the latent dimension was 128, the triplet loss margin was 0.05, the negative selection type was ‘semihard’, the triplet loss weight was 0.001, the learning rate was 0.005, the batch size was 256, the $T_{\max}$ for cosine annealing was 0, the dropout was set to 0.5, the input dropout was set to 0.4, the L_1 -regularization penalty was 0.0001 and the L_2 -regularization penalty was 0.01. The amount of training computation was normalized for all models to a batch size of 1,000, with 100 batches per epoch and a maximum of 500 epochs (when pretraining SCimilarity, an epoch is defined not as a pass of the full dataset but as an arbitrary number of batches used to form triplets).

Note on preprocessing datasets for each model. The software for scVI takes in raw gene expression counts. As a result, we did not perform any preprocessing on the pretraining or evaluation data before applying scVI. Similarly, the Geneformer software requires raw gene expression counts before it tokenizes the data; thus, we did not perform any preprocessing before this step in the model. For Pretrained PCA, SSL and SCimilarity, gene expression profiles were normalized to 10,000 counts per cell and then logarithmized with a pseudocount of 1 (that is, processed with the function $f(\mathbf{x}) = \log(\mathbf{x} + 1)$).

Experiments assessing the impact of model size

To assess the impact of model size on model performance, we performed two sets of model scaling experiments using scVI (because it was the best performing deep generative model in the zero-shot evaluations) and Geneformer (as a representative transformer-based model). To scale scVI, we modulated both the number of nodes per hidden layer and the dimension of the latent embedding space. Note that we kept all other parameters and hyperparameters the same as described above. This generated models that had approximately 0.5×, 2×, 4×, 8× and 16× the number of parameters as the default scVI implementation that we highlight in the main text. To scale Geneformer, we modulated the latent embedding space from a default value of 256 to additional values of 32, 64 and 128.

Experiments assessing the impact of hyperparameter settings

In the next set of experiments, we assessed the impact of different hyperparameter settings on model downstream performance. First, to investigate the influence of weight decay on Geneformer's downstream performance, we pretrained different versions of Geneformer, as described above, but with additional weight decay values of 0.01 and 0.0001 (note that the default used was 0.001). Next, we investigated the influence of different training strategies in autoencoder-based models. Both scVI and SSL are autoencoders, where the former uses a variational loss function and the latter makes use of a masked pretraining objective. To assess the relative effects of both approaches, we additionally pretrained SSL using both the masked training objective and a variational loss term.

Description of model fine-tuning for cell-type classification

Below is a description of how we fine-tune on the basis of the pretrained PCA, scVI, SSL, Geneformer and SCimilarity models.

Pretrained PCA, scVI and SCimilarity. These three models do not have a fine-tuning mechanism built-in. To get around this fact, we implemented a simple MLP that utilized the model embeddings from each of these models as input (this can be conceptualized as freezing the weights of the encoder for each of these models, where PCA is viewed as a linear autoencoder). We utilized an MLP that closely matched the architecture of the scVI decoder, but whose output predicted the cell-type label rather than reconstructing gene expression. The model architecture was an input layer of the size of the latent embedding space for the corresponding pretrained model, a hidden layer of 128 neurons, a ReLU function after the hidden layer and an output linear layer the size of the number of cell-type labels. This architecture was chosen to be simple and was not hyperparameter optimized. For training, we used a dropout rate of 0.2, a batch size of 128, the Adam optimizer with a learning rate of 1×10^{-4} , cross-entropy loss, early stopping based on the validation loss and 500 maximum epochs. PCA, scVI and SCimilarity were trained on heterogeneous hardware consisting of both central processing units and various types of GPUs.

SSL. To fine-tune SSL, we used the built-in cell-type classification model from the SSL library. We used the default model architecture where the MLP hidden layers had sizes of 512, 512, 256, 256 and 64. For SSL, no

weights were frozen during fine-tuning, and for training, we used the AdamW optimizer with a learning rate of 9×10^{-4} , a dropout rate of 0.1, a weight decay of 0.05, a batch size of 16,384, a minimum of 30 epochs, used early stopping based on validation loss and the same number of maximum epochs used during pretraining on a single NVIDIA 80 GB A100 GPU. The learning rate and weight decay were nondefault values and were taken from the SSL training scripts.

Geneformer. To fine-tune Geneformer, we used the built-in cell-type classification model from the Geneformer library. We froze the first two layers of the pretrained model. The fine-tuned MLP classifier adds a final, classification transformer layer. For training, we used a single epoch (following the example set by Theodoris et al.¹³), with 1,000 warm-up steps, a linear learning rate schedule, a batch size of 18 per device and eight NVIDIA 80 GB A100 GPUs. We used a learning rate of 0.000210031 and a weight decay of 0.233414 as obtained by using Geneformer's hyperparameter optimizer.

Classification tasks and evaluation metrics

We used two simple embeddings as baselines for zero-shot cell-type classification: (1) HVGs and (2) the cell loadings of the first 50 principal components computed on the dataset. As a baseline for fine-tuned cell-type classification, we trained a regularized logistic classifier on the HVGs determined on the training dataset. In both cases, for a classification task with cell-type labels $\mathbf{y} = (y_1, \dots, y_N)$ and predicted cell types $\hat{\mathbf{y}} = (\hat{y}_1, \dots, \hat{y}_N)$, we use the following metrics: accuracy, precision, recall, macro F1 score and micro F1 score.

Accuracy measures the fraction of cells that are correctly predicted

$$\text{Accuracy}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\hat{y}_i = y_i), \quad (1)$$

where $\mathbb{1}(\cdot)$ is an indicator function. For a classification problem, precision and recall are defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \text{ and } \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (2)$$

where TP denotes the number of true positives, FP is the number of false positives and FN is the number of false negatives. Last, the F1 score is the harmonic mean of the precision and recall such that

$$\text{F1} = \frac{\text{TP}}{\text{TP} + (\text{FP} + \text{FN})/2}, \quad (3)$$

where, in addition to previous notation, TN is the number of true negatives. In this study, we assessed performance in datasets with multiple cell types. There are two variants of the F1 score that measure performance when there are more than two classes. The micro F1 score computes TP, FN and FP globally across all cell types. The macro F1 score computes the binary F1 score for each individual label and then averages these F1 scores (which weights each cell type equally, regardless of if they are present in unequal proportions).

Batch integration task and evaluation metrics

We used two simple embeddings as baselines for zero-shot batch integration: (1) HVGs and (2) the cell loadings of the first 50 principal components computed on the dataset. For an integration task with cell-type labels $\mathbf{y} = (y_1, \dots, y_N)$, model embeddings are used to obtain cluster labels $\hat{\mathbf{y}} = (\hat{y}_1, \dots, \hat{y}_N)$. Here, we use the following metrics: NMI (cluster/label), ARI (cluster/label), ASW (label), ASW (batch) and AvgBIO as implemented by Luecken et al.³⁹ in the scib package.

NMI. For two sets of cell-type labels, the NMI captures the similarity between the two labels. In the zero-shot integration task, the two labels

are the annotated cell-type assignments and the clusters derived from the model embeddings. The NMI is computed as follows:

$$\text{NMI}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{2I(\mathbf{y}, \hat{\mathbf{y}})}{H(\mathbf{y}) + H(\hat{\mathbf{y}})}, \quad (4)$$

where $I(\mathbf{y}, \hat{\mathbf{y}})$ is the mutual information of $\mathbf{y}$ and $\hat{\mathbf{y}}$, and $H(\bullet)$ is the Shannon entropy.

ARI. For two sets of cell-type labels, the ARI captures the similarity between the two labels. In the zero-shot integration task, the two labels are the annotated cell-type assignments and the clusters derived from the model embeddings. It is based on the Rand Index (RI) which is computed as the follows:

$$\text{RI} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}. \quad (5)$$

The ARI corrects for the RI measurement's sensitivity to chance via permutation⁶¹ where

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{1 - \mathbb{E}[\text{RI}]}. \quad (6)$$

Here, $\mathbb{E}[\text{RI}]$ denotes the expectation of the RI.

ASW. We compute the ASW considering cell-type labels (ASW_L), to measure how well embeddings are retaining biological information. We then normalize this score via the following

$$\text{ASW}_{\text{label}} = \frac{\text{ASW}_L + 1}{2}. \quad (7)$$

The metric $\text{ASW}_{\text{label}}$ ranges from 0 to 1, where higher scores indicate better cell-type clustering. We also compute the ASW considering batch labels (ASW_B), to measure how well embeddings are retaining batch effects. We then invert this score to measure how well embeddings are removing batch effects where

$$\text{ASW}_{\text{batch}} = 1 - |\text{ASW}_B|. \quad (8)$$

The metric $\text{ASW}_{\text{batch}}$ also ranges from 0 to 1, where higher scores indicate better batch-mixing performance.

AvgBIO. The AvgBIO score is the arithmetic mean of ASW, NMI and ARI where

$$\text{AvgBIO} = \text{ARI}_{\text{cell}} + \text{NMI}_{\text{cell}} + \text{ASW}_{\text{cell}}. \quad (9)$$

This metric was introduced by Cui et al.¹⁴ and is intended to measure the biological conservation of the embeddings being evaluated. AvgBIO is normalized to the unit scale, with 1 indicating complete agreement between clusters and cell-type labels.

Perturbation response prediction task and evaluation metrics

To perform fine-tuned perturbation response prediction, we implemented the following strategy motivated by Lotfollahi et al.⁴. First, we pretrained PCA, scVI, SSL, Geneformer and SCimilarity on some set of unperturbed cells from scTab. During fine-tuning, we took the embeddings corresponding to an unperturbed cell and trained a simple MLP to predict the gene expression of a real (and randomly selected) perturbed cell of the same type. The MLP had an architecture with an input layer the size of each model's embedding dimension (that is, 50 for pretrained PCA, 10 for scVI, 64 for SSL, 256 for Geneformer and 128 for SCimilarity), a hidden layer of 128 neurons, a ReLU function after

the hidden layer and an output linear layer size equal to the number of cell-type labels. Note that this architecture was chosen to be simple and was not optimized. During fine-tuning, we used a dropout rate of 0.2, a batch size of 128, an Adam optimizer with a learning rate of 1×10^{-4} , cross-entropy loss, early stopping based on the validation loss and 500 maximum epochs.

When conducting evaluation, we took embeddings corresponding to unseen unperturbed cells from a particular cell line and generated a predicted gene expression vector using the fine-tuned MLP. We then compared these predictions with the average expression of real perturbed cells from the same cell line (on a per-gene basis). We used two metrics to assess perturbation response prediction accuracy: (1) the coefficient of determination (or R^2) and (2) MSE. All R^2 values were calculated by squaring the r output from the `scipy.stats.linregress` function.

Data diversity measures

We used three different diversity measures to assess the impact of the various downsampling schemes on the diversity of the data used for model pretraining. The Shannon Index and Gini–Simpson Index are based on cell metadata, while the Vendi Score is based on the gene expression values directly. For the two metadata-based measures, we use the following notation: let $p = (p_1, \dots, p_K)$ be the proportion of each cell type in the dataset where K is the total number of cell types.

Shannon Index. The Shannon Index is a popular diversity measure in ecology³⁰ and is the application of Shannon entropy²⁹ to group proportions. In particular, the Shannon Index is given by

$$H' = -\sum_i p_i \log_2 p_i. \quad (10)$$

While the Shannon Index can be defined using any base for the logarithm, in this study we use base 2. For the Shannon Index, a higher value indicates increased diversity.

Simpson Index and Gini–Simpson Index. The Simpson Index is also a popular diversity index in ecology⁶² and represents the probability that two randomly chosen cells belong to the same cell type

$$\gamma = \sum_i p_i^2. \quad (11)$$

For the Simpson Index, a higher value (counter-intuitively) indicates decreased diversity. This leads to the definition of the Gini–Simpson Index, which is simply the complement of the Simpson index

$$\text{Gini–Simpson Index} = 1 - \gamma \quad (12)$$

For the Gini–Simpson Index, a higher value indicates increased diversity.

Vendi Score. The Vendi Score is a measure of the intrinsic diversity of a dataset, without reference to any metadata³¹. In this study, we calculate the Vendi Score of the gene expression matrix for each dataset. When using a linear kernel, the Vendi Score has computational complexity $O(\min(J^2, N^3))$ where N is the number of samples in the dataset and J is the number of features in the dataset. Owing to this, we preprocessed each dataset by:

1. Normalizing cells to 10,000 counts per cell.
2. Logarithmizing the counts with a pseudocount of 1 (that is, using the function $f(\mathbf{x}) = \log(\mathbf{x} + 1)$).
3. Restricting to HVGs using the scanpy function `sc.pp.highly_variable_genes` with default parameters.
4. Centering and scaling the variable genes that remained.

Suppose that we have a collection of cells $(\mathbf{x}_1, \dots, \mathbf{x}_N)$ and define a positive semidefinite similarity function k that satisfies $k(\mathbf{x}, \mathbf{x}) = 1$.

Next let K be an $N \times N$ kernel matrix where each entry is $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$, and let $\lambda_1, \dots, \lambda_N$ be the eigenvalues of K/N . The Vendi Score with respect to the function k is then the exponential of the Shannon entropy of the eigenvalues of K/N where

$$VS_k(\mathbf{x}_1, \dots, \mathbf{x}_N) = \exp\left(-\sum_i \lambda_i \log \lambda_i\right). \quad (13)$$

The Vendi Score can also be viewed as the effective rank of a matrix and was previously introduced in the signal processing literature⁶³.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The clonal hematopoiesis, placenta, intestine-on-chip, periodontitis, renal and liver datasets are all available via the CELLxGENE census at <https://cellxgene.cziscience.com/>. The lung dataset from Kim et al.³⁶ is available via the Curated Cancer Cell Atlas at <https://www.weizmann.ac.il/sites/3CA/lung>. The Tahoe-100M dataset is available online via Google Cloud Storage (under bucket gs://arc-ctc-tahoe100/2025-02-25/) and instructions for downloading can be found at <https://github.com/ArcInstitute/arc-virtual-cell-atlas/blob/main/tahoe-100M/tutorial-py.ipynb>. Instructions for downloading the scTab corpus from the CELLxGENE census are available via GitHub at <https://github.com/microsoft/scFM-dataselection>, as well as via Zenodo at <https://doi.org/10.5281/zenodo.19173545> (ref. 64). Source data are provided with this paper.

Code availability

All code is available under an open-source MIT license available via GitHub at <https://github.com/microsoft/scFM-dataselection>, as well as via Zenodo at <https://doi.org/10.5281/zenodo.19173545> (ref. 64). Specifically, we used scVI¹, which is available via scvi-tools at <https://scvi-tools.org/> ref. 65 (release 1.1.2), the version of SSL²⁵ corresponding to commit a149b403fb2f7fe976cb8bd30fcb3db7a51e9500 (2 May 2024) is available via GitHub at https://github.com/theislabs/ssl_in_scg, the version of Geneformer¹³ corresponding to commit b2bbd7cc856e3b3c1a9199c7cca07d888d87663 (29 June 2024) is available via the Hugging Face repository at <https://huggingface.co/ctheodoris/Geneformer/tree/main/geneformer> and SCimilarity (release 0.4.1) is available via GitHub at <https://github.com/Genentech/scimilarity>.

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Acknowledgements

This research was conducted using a combination of computational resources and services provided by Microsoft Research and the Center for Computation and Visualization at Brown University. This research was also supported in part by a David & Lucile Packard Fellowship for Science and Engineering awarded to L.C. S.R. acknowledges funding support from NCI K08 CA260442. Any opinions, findings and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of any of the funders.

Author contributions

A.D., A.P.A. and L.C. conceived the study. A.D., M.H. and A.T. developed the software and performed the analyses. A.G. and A.W.N. conducted secondary analyses. N.F., S.R., P.S.W., A.P.A. and L.C. provided resources. S.R., P.S.W., A.P.A. and L.C. supervised the project. A.D., M.H., A.T., A.P.A. and L.C. wrote the initial draft. All authors interpreted the results and revised the paper.

Competing interests

S.R. holds equity in Amgen. P.S.W. reports compensation for consulting/speaking from Engine Ventures, AbbVie and Passkey Therapeutics unrelated to this work. S.R. and P.S.W. receive research funding from Microsoft. M.H., N.F., A.P.A. and L.C. are employees of Microsoft and own equity in Microsoft. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41592-026-03120-y>.

Correspondence and requests for materials should be addressed to Ava P. Amini or Lorin Crawford.

Peer review information *Nature Methods* thanks Felix Fischer, Fabian Theis and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available. Primary Handling Editor: Lin Tang, in collaboration with the *Nature Methods* team.

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Data collection	No software was used to collect data.
Data analysis	All code is available under an open-source MIT license at https://github.com/microsoft/scFM-dataselection . Specifically, we used scVI which was downloaded from https://scvi-tools.org/ (release 1.1.2), the version of SSL corresponding to commit a149b403fb2f7fe976cb8bd30fcb3db7a51e9500 (May 2, 2024) from the GitHub repository https://github.com/theislab/ssl_in_scg , the version of Geneformer corresponding to commit b2bbd7ccc856e3b3c1a9199c7cca07d888d87663 (Jun 29, 2024) from the Hugging Face repository https://huggingface.co/ctheodoris/Geneformer/tree/main/geneformer , and SCimilarity (release 0.4.1) from https://github.com/Genentech/scimilarity .

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Data exclusions	There was no exclusion of data.
Replication	Not applicable. We used publicly available datasets and did not have control over replication in the study design.
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